

HARSHIT AGARWAL

+91 9883830998 harshitagrawal5160@gmail.com GitHub LinkedIn LeetCode CodeChef Portfolio

EDUCATION

Jalpaiguri Government Engineering College, West Bengal

Bachelor of Technology in Electronics and Communication Engineering (ECE)

Aug 2024 – Present

YGPA: 8.16

EXPERIENCE

Indian Institute of Technology, Guwahati

Research Intern

Jun 2026 – Jul 2026

Guwahati, Assam

- Developed **HelioSense**, an end to end deep learning framework for forecasting solar flares using **ISRO Aditya-L1 (SoLEXS/HELIOS)** and NASA GOES satellite observations.
- Engineered a scalable satellite data pipeline through FITS parsing, GTI based quality validation, temporal synchronization, and multivariate feature generation for deep learning.
- Designed a hybrid **1D CNN + PatchTST** architecture that captures local temporal patterns and long-range dependencies in multivariate solar X-ray time-series data for accurate flare prediction.
- Implemented a multi-task learning framework for simultaneous solar flare nowcasting and forecasting, achieving **83.6% nowcast accuracy** while predicting future flare occurrence, peak X-ray flux, and **6-hour flare probability** up to a **3-hour forecasting horizon**.

PROJECTS

campAIgn: AI Marketing Campaign Automation Platform

React, Node.js, MongoDB, n8n, OpenAI API

- Deployed a comprehensive email marketing solution, enhancing campaign creation and preview capabilities, resulting in over **500** successful campaigns executed through the platform.
- Designed high-performance **REST APIs** with authentication (**JWT**) ensuring secure synchronization of event-driven automation workflow triggers.
- Developed event-driven automation workflows using **n8n** to handle campaign generation, audience segmentation, and email delivery.
- Implemented customer segmentation pipeline using **Python (Scikit-Learn)** to process **5,000+** records and enable targeted messaging.

Sentiment Classification from Text

Python, scikit-learn, NLTK, TF-IDF, Streamlit

- Built a 6-class emotion classifier (joy, sadness, anger, fear, love, surprise) on **16,000+** sentences using **TF-IDF bigrams** and **Logistic Regression**, achieving 90% test accuracy.
- Benchmarked 3 approaches (**BoW+NaiveBayes**, **TF-IDF+NaiveBayes**, **TF-IDF+LR**) and selected the best-performing pipeline to handle class imbalance.
- Deployed an interactive **Streamlit** app with real-time emotion prediction and per-class confidence scores.

TECHNICAL SKILLS

- Languages:** Java, Python, C
- Core CS:** Data Structures and Algorithms (DSA), OOPs, OS, CN, DBMS
- Web Development:** HTML, CSS, JavaScript, Node.js, Express.js, EJS
- Machine Learning & Deep Learning:** ANN, CNN, NLP, RNN (LSTM, GRU), Transformers (BERT, ELMO), LLM Fine Tuning, Token Classification, Computer Vision
- Libraries & Frameworks:** Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, TensorFlow, Keras
- Tools:** Git, GitHub, VS Code, Jupyter Notebook, Google Colab, Excel, Jupyter Lab

ACHIEVEMENTS

- LeetCode:** Solved **400+** DSA problems on LeetCode and various coding platforms
- CodeChef:** Achieved 2 star rating on CodeChef (Max Rating: 1414)
- Research:** Published **4 Research Papers** accepted at **Springer Nature** and **IEEE** in Data Driven ML Applications
- Awarded a merit-cum-means scholarship by the Government of West Bengal
- Completed multiple online **Certifications** in relevant fields